

ABHINESH LODHI

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PROFILE SUMMARY

ML Engineer with experience building multimodal AI systems, transformer architectures, model training, fine-tuning, dataset engineering, and deployment pipelines using Python, PyTorch, TensorFlow, Docker, and MLflow.

EDUCATION

PG-DBDA (Post Graduate Diploma in Big Data Analytics) Aug 2025 – Feb 2026
C-DAC, Bengaluru

Bachelor of Technology (Computer Science Engineering) 2020 – 2024
Technocrats Institute of Technology & Science, Bhopal

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, JavaScript, Bash
- **AI/ML:** Machine Learning, Deep Learning, NLP, Transformers, Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face, Pandas, NumPy, Feature Engineering, Model Training, Model Evaluation, Predictive Modeling, Supervised & Unsupervised Learning, Classification, Regression, Clustering, Ensemble Methods, Cross Validation, Accuracy, Precision, Recall, F1-Score, RMSE, AUC
- **Generative AI & LLMs:** LangChain, LangGraph, LlamaIndex, OpenAI APIs, RAG, Prompt Engineering, AI Agents, Semantic Search, Embeddings, Vector Databases, FAISS, Pinecone
- **Backend & Data Engineering:** Django, FastAPI, Flask, REST APIs, ETL Pipelines, Workflow Automation, Apache Spark, Hadoop, Kafka, Airflow
- **Databases & Tools:** MySQL, Redis, Elasticsearch, Git, GitHub, Linux, Docker, MLflow, DVC, AWS Fundamentals, OCI, Jupyter Notebook, Google Colab, VS Code
- **Computer Vision & Multimodal AI:** OCR, Computer Vision, Multimodal Models, Vision Transformers (ViT), Image Processing, Document Understanding, Unstructured Data Processing
- **MLOps & Deployment:** Docker, MLflow, Model Deployment, Kubernetes, Experiment Tracking, CI/CD, Containerization, End-to-End ML Pipelines
- **Research & Evaluation:** Dataset Curation, Benchmarking, Hyperparameter Tuning, Model Fine-tuning, Accuracy Optimization, Robustness Testing, Performance Evaluation

EXPERIENCE

Python Intern Jun 2025 – Aug 2025
CredenTek Software & Consultancy Pvt. Ltd., Pune, India

- Built Python-based data ingestion and preprocessing pipelines for structured and semi-structured enterprise datasets, improving validation, transformation, and workflow automation.
- Automated dataset processing workflows supporting scalable data engineering operations using Python and Agile SDLC practices.
- Developed backend processing systems for enterprise data exchange platforms emphasizing reliability, validation, and operational efficiency.

PROJECTS

Multi-Agent AI Research Assistant

Tech Stack: Python, FastAPI, LangGraph, LangChain, Hugging Face, FAISS, Pinecone, Redis, MLflow, QLoRA, Docker

- Fine-tuned transformer-based models using Hugging Face and QLoRA techniques, improving contextual understanding, semantic retrieval quality, and inference efficiency.
- Implemented ML experimentation workflows using MLflow, Docker containerization, and evaluation pipelines for benchmarking model performance.
- Built scalable retrieval and orchestration systems using vector databases, memory layers, and autonomous multi-agent workflows.

Multimodal Document Understanding System

PyTorch, HuggingFace, OCR, LayoutLMv3, OpenCV, MLflow, Docker

- Built multimodal document parsing pipeline combining OCR, layout-aware transformers, and computer vision for extracting structured information from PDFs.
- Fine-tuned LayoutLM-based architectures on custom annotated datasets for document understanding tasks.
- Benchmarked model performance using accuracy and robustness metrics while tracking experiments through MLflow.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 - AI Foundations Associate (Credential)